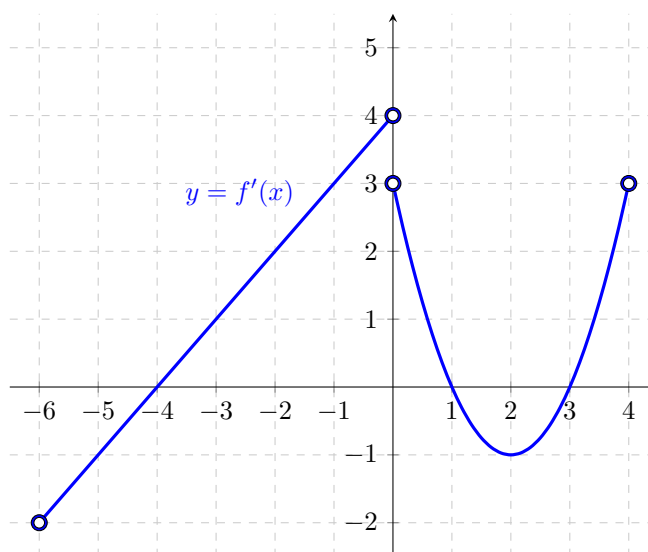


Problem 5

Problem 5

Problem 1 Let f be a function that is continuous on its domain $(-6, 4)$. The graph of f' , the derivative of f , is given below.



Based on the graph of f' , answer the question below.

- (a) List x-coordinates of all critical points of f .

Explanation. Notice, $f'(-4) = f'(1) = f'(3) = 0$, and $f'(0)$ is not defined. Therefore, the function f has critical points at $x = -4$, $x = 0$, $x = 1$ and $x = 3$.

- (b) List x-coordinates of all critical points of f where $f'(x) = 0$.

Explanation. $x = -4$, $x = 1$ and $x = 3$.

- (c) List x-coordinates of all critical points of f where $f'(x)$ is **undefined**.

Explanation. $x = 0$

- (d) List x-coordinates of all local minimums of f .

Problem 5

Explanation. $x = -4$ and $x = 3$. At these points the sign of $f'(x)$ changes from negative to positive.

- (e) List x-coordinates of all local maximums of f .

Explanation. $x = 1$ At this point the sign of $f'(x)$ changes from positive to negative.

- (f) List all intervals where f is decreasing and concave down.

Explanation. $(1, 2)$ Notice, $f'(x)$ is negative and decreasing there.

- (g) List all intervals where f is decreasing and concave up.

Explanation. $(-6, -4)$ and $(2, 3)$. Notice, $f'(x)$ is negative and increasing there.

- (h) List all intervals where f is increasing and concave down.

Explanation. $(0, 1)$ Notice, $f'(x)$ is positive and decreasing there.

- (i) List all intervals where f is increasing and concave up.

Explanation. $(-4, 0)$ and $(3, 4)$. Notice, $f'(x)$ is positive and increasing there.

- (j) List x-coordinates of all inflection points of f .

Explanation. $x = 0$, and $x = 2$ $f'(x)$ is increasing on $(-6, 0)$ and decreasing on $(0, 2)$, then increasing on $(2, 4)$.